

TABLE 1 Characteristics of insertion sequence families

Families	Sub-groups	Typical size-range	DR (bp)	Ends	IRs	No. of ORF	Frameshift	Catalytic residues	Comments	Mechanism
IS1	-	740–1,180	8–9	GGnnnTG	Y	2	ORFAB	DDE		copy-and-paste and cointegrate
	single ORF	800–1,200	0–9			1				
	ISMhu11	900–4,600	0–10		Y	2	ORFAB			
IS1595	ISPna2	1,000–1,150	8	GGCnnTG	Y	1		DDNK		copy-and-paste ?
	ISPna2+pass	1,500–2,600	8							
	ISH4	1,000	8	CGCTCTT		1		DDNK		
	IS1016	700–745	7–9	GGGgctg		1		DDEK		
	IS1595	900–1,100	8	CcTGATT		1		DDNK + ER4R7		
	ISSod11	1,000–1,100	8	nnnGcnTATC		1		DDHK + ER4R7		
	ISNwi1	1,080–1,200	8	ggnnatTAT		1		DDEK + ER4		
	ISNwi1+pass	1,750–4,750	8			1				
	ISNha5	3,450–7,900	8	CGGnnTT		1		DDER/K		
IS3	IS150	1,200–1,600	3–4	TG	Y	2	ORFAB	DDE		copy-paste
	IS407	1,100–1,400	4	TG						
	IS51	1,000–1,400	3–4	TG						
	IS3	1,150–1,750	3–4	TGa/g						
	IS2	1,300–1,400	5	TG						
IS481	-	950–1,300	4–15	TGT	Y	1		DDE		copy-paste ?
ISNCY	IS1202	1,400–1,700	5	TGT	Y	1		DDEQ		
IS4	IS10	1,200–1,350	9	CT	Y	1		DDE	Hairpin intermediate	cut-and-paste
	IS50	1,350–1,550	8–9	C					Hairpin intermediate	
	ISPep1	1,500–1,600	7–8	-T-AA					?	
	IS4	1,400–1,600	10–13	-AAT					?	
	IS4Sa	1,150–1,750	8–10	CA					?	
	ISH8	1,400–1,800	10	CAT					?	
	IS231	1,450–5,400	10–12	CAT		1 or + *			* Passenger genes	
IS701	-	1,400–1,550	4		Y	1		DDE		
	ISAbal1				Y	1		DDE		
ISH3	-	1,225–1,500	4–5	C-GT	Y	1		DDE		
IS1634	-	1,500–2,000	5–6	C	Y	1		DDE		
IS5	IS903	950–1,150	9	GG	Y	1		DDE		
	ISL2	850–1,200	2–3			1				
	ISH1	900–1,150	8	-GC		1				
	IS5	1,000–1,500	4	Ga/g		1				
	IS1031	850–1,050	3	GAa/g		1				
	IS427	800–1,000	2–4	Ga/g		2	ORFAB			

IS1182	-	1,330–1,950	0–60		Y	1		DDE	
ISNCY	ISDo1	1,600–1,900	6–7		Y	1		DDE	
IS6	-	700–900	8	GG	Y	1		DDE	co-integrate
IS21	-	1,750–2,600	4–8	TG	Y	2 *		DDE	* istB: transposition helper
IS30	-	1,000–1,700	2–3		Y	1		DDE	copy-and-paste
IS66	-	2,000–3,000	8–9	GTAA	Y	3 *		DDE *	* TnpC has the DDE domain
	ISBst12	1,350–1,900	8–9	GTAA	Y	1		DDE	
IS256	-	1,200–1,500	8–9	Ga/g	Y	1		DDE	copy-paste
	IS1249	1,300	0–10	GG	Y	1		DDE	
	ISC1250	1,250	0–9	GG	Y	1		DDE	
ISH6	-	1,450	8	GGT	Y	1		DDE	
ISLre2	-	1,500–2,000	9		Y	1		DDE	
ISKra4	ISAzba1	1,400–2,900	0		Y	1 or + *		DDE	* Passenger genes
	ISMicb2	1,250–1,400	8	GGG	Y	1 or 2	ORFAB	DDE	
	ISKra4	1,400–3,700	9	GGG	Y	1 or + *		DDE	* Passenger genes
IS630	-	1,000–1,400	2 *		Y	1 or 2	ORFAB	DDE	* Target site : often NTAN with duplication of the TA cut-and-paste
IS982	-	1,000	3–9	AC	Y	1		DDE	
IS1380	-	1,550–2,000	4–5	CC	Y	1		DDE	
ISAs1	-	1,200–1,500	8–10	CAGGG	Y	1			
ISL3	-	1,300–2,300	8	GG	Y	1			
Tn3	-	>3,000	0	GGGG	Y	>1		DDE	co-integrate
ISAz13	-	1,250–2,200	0–4	Ga/g	Y	1			
IS110	-	1,200–1,550	0		N	1		DEDD	* IRs not at the termini of the IS
	IS1111				Y *				
IS91	-	1,500–2,000	0		N	1		HUH/Y2	Target site GAAC () CAAG rolling circle
IS200/IS605	IS200	600–750	0		N	1 *		HUH/Y1	* TnpA * TnpA + TnpB ; ** Y1 on TnpA * TnpB peel-and-paste
	IS605	1,300–2,000				2 *		HUH/Y1 **	
	IS1341	1,200–1,500				1 *			

IS607	-	1,700–2,500	0		N	2 *		Serine **	* TnpA + TnpB; ** TnpA
ISNCY	IS892	1,600	0–8	CTAG	Y	2	ORFAB		
	ISLbi1	1,400–1,500	5		Y	1			
	ISMae2	1,400–2,400	9	CAG	Y	1			
	ISPlu15	800–1,000	0		N	1			
	ISA1214	1,000–1,200	8–12		Y	2			
	ISC1217	1,200	6–8	TAG	Y	1			
	ISM1	1,300–1,600	8–9		Y	1			

Abbreviations: DR, duplication repeat; IS, insertion sequence; ORF, open reading frame.